

MICROSOFT SENTINEL & DEFENDER

SOC Detection, Threat Hunting & Incident Response Lab

Brute Force Attack Simulation | KQL Threat Hunting | Custom Analytics Rule Creation | Incident Detection, Response & Remediation

Resource Group: Security-BikSecOps (Australia East)
Victim VM: Web01 -- Windows Server 2022 Datacenter (Public IP: 20.5.48.127)
Attacker VM: AttackerVM -- Ubuntu 24.04 LTS
Log Analytics Workspace: Bik-SOC-Lab
SIEM Platform: Microsoft Sentinel + Microsoft Defender XDR (Unified Portal)
Attack Simulated: RDP Brute Force using Hydra (from AttackerVM)
Detection Tool: KQL -- SecurityEvent (Event IDs 4624, 4625)
Threat Hunting: Queried Sentinel Logs using KQL -- confirmed failed RDP attempts from attacker IP followed by 1 successful login for user bikashsoc1
Analytics Rule: Successful Brute Force Attempt (High severity)
MITRE ATT&CK: T1110 Brute Force | T1110.001 Password Guessing | T1078 Valid Accounts
Incident ID: ID 18 -- Successful Brute Force Attempt
Incident Response: Triaged Incident ID 18 in Defender XDR, assigned investigation task (User/Host Review), reviewed authentication logs, documented containment actions -- simulated full IR lifecycle including Containment, Eradication, Recovery and Escalation steps
Date: June 2026
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1. Lab Overview

This lab demonstrates a complete SOC workflow using Microsoft Sentinel and the Microsoft Defender XDR unified portal. The lab covers infrastructure deployment, attack simulation, log analysis using KQL, custom analytics rule creation, incident detection, and incident response -- replicating real-world SOC operations.

A Windows Server VM (Web01) was intentionally exposed with an open NSG inbound rule and Windows Firewall disabled to simulate a vulnerable internet-facing server. An Ubuntu attacker VM used Hydra to launch a dictionary-based RDP brute force attack, successfully authenticating after multiple failed attempts. Microsoft Sentinel detected the attack via a custom KQL analytics rule, triggered a High-severity incident in the Defender XDR portal, and the incident was investigated and closed following simulated SOC procedures.

2. Lab Objectives

- * Deploy a resource group, Windows Server VM (victim), and Ubuntu VM (attacker) in Azure.
- * Configure a vulnerable NSG inbound rule (Any Any *) and disable Windows Defender Firewall on Web01.
- * Create a Log Analytics Workspace (Bik-SOC-Lab) and onboard Microsoft Sentinel.
- * Install Windows Security Events via AMA connector and create a Data Collection Rule.
- * Launch an RDP brute force attack from AttackerVM using Hydra and a common password list.
- * Verify brute force activity in Sentinel Logs using KQL (Event IDs 4624 and 4625).
- * Create a scheduled analytics detection rule mapped to MITRE ATT&CK T1110.
- * Validate incident creation in Microsoft Defender XDR unified portal.
- * Perform simulated incident response: assign task, investigate, document, and close incident.

3. Lab Environment

Resource Group	Security-BikSecOps
Region	Australia East (West Pacific)
Victim VM	Web01 -- Windows Server 2022 Datacenter Standard D2s v3
Victim Public IP	20.5.48.127
Victim Private IP	10.0.0.4
NSG Rule	Inbound -- Any Source / Any Destination / Any Port / Allow (Named: Dangerous)
Windows Firewall	Turned OFF on Web01 (Public and Private profiles)
Attacker VM	AttackerVM -- Ubuntu 24.04 LTS
Attack Tool	Hydra v9.5 (THC-Hydra) -- RDP brute force
Log Analytics Workspace	Bik-SOC-Lab (australiaeast)
SIEM	Microsoft Sentinel
Unified Portal	Microsoft Defender XDR (security.microsoft.com)
Data Connector	Windows Security Events via AMA
DCR Name	SecurityEventsforAzureVM -- All Events

4. Phase 1 -- Infrastructure Deployment

4.1 Create Resource Group

A dedicated Resource Group was created to contain all lab resources.

Resource Group Name	Security-BikSecOps
Subscription	Azure Subscription 1
Region	Australia East (West Pacific)

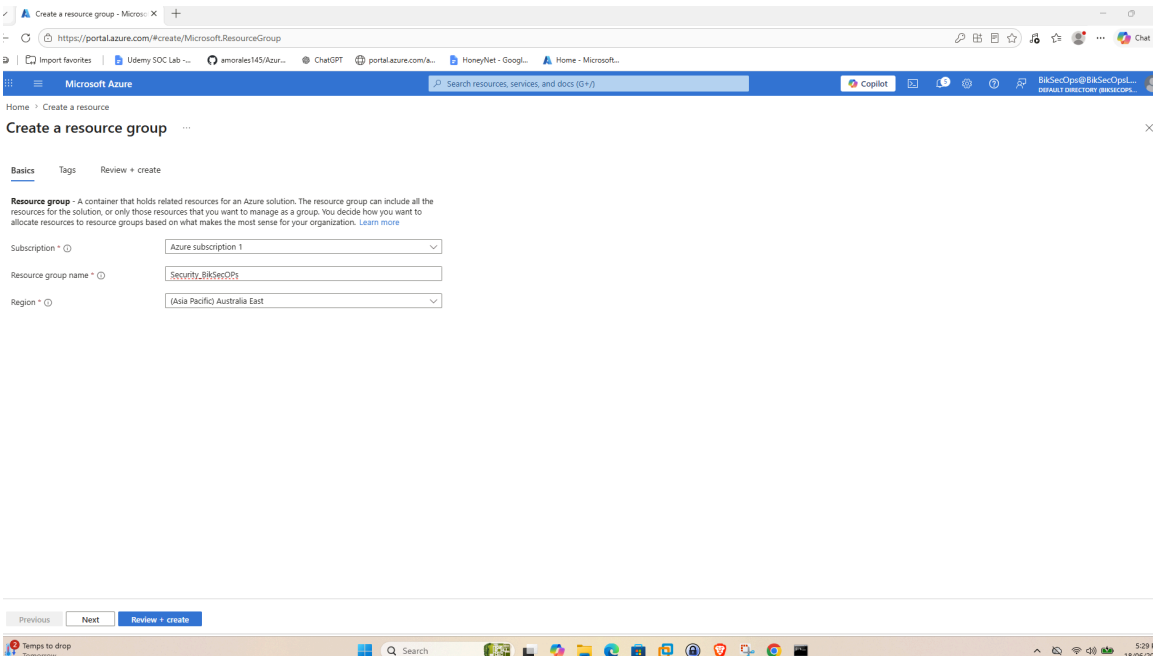


Fig 1 -- Azure Portal: Create resource group -- Security-BikSecOps, Australia East, Review+create

4.2 Create Victim VM -- Web01 (Windows Server 2022)

A Windows Server 2022 Datacenter VM was created with a deliberately insecure NSG inbound rule to simulate an exposed internet-facing server.

VM Name	Web01
OS	Windows Server 2022 Datacenter
Size	Standard D2s v3 (2 vCPUs, 8 GB RAM)
Public IP	20.5.48.127
Private IP	10.0.0.4
NSG Inbound Rule	Source: Any / Destination: Any / Port: * / Protocol: Any / Action: Allow / Name: Dangerous

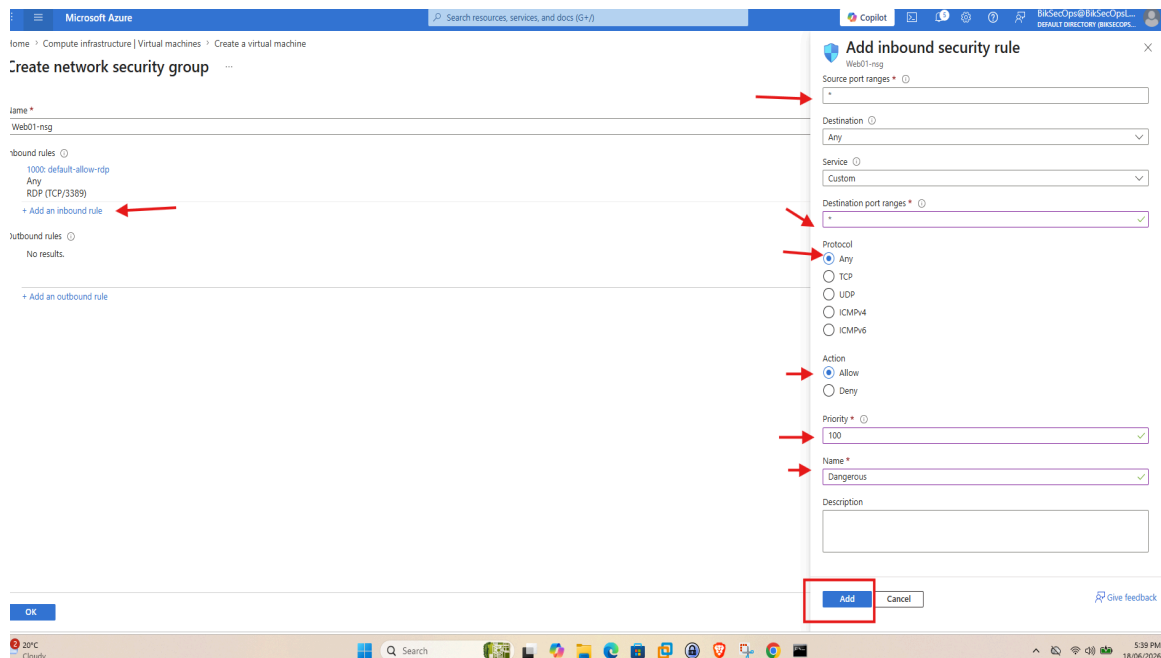


Fig 2 -- Azure: Create NSG inbound rule -- Any/Any/* Allow (Priority 100, Name: Dangerous) to expose Web01

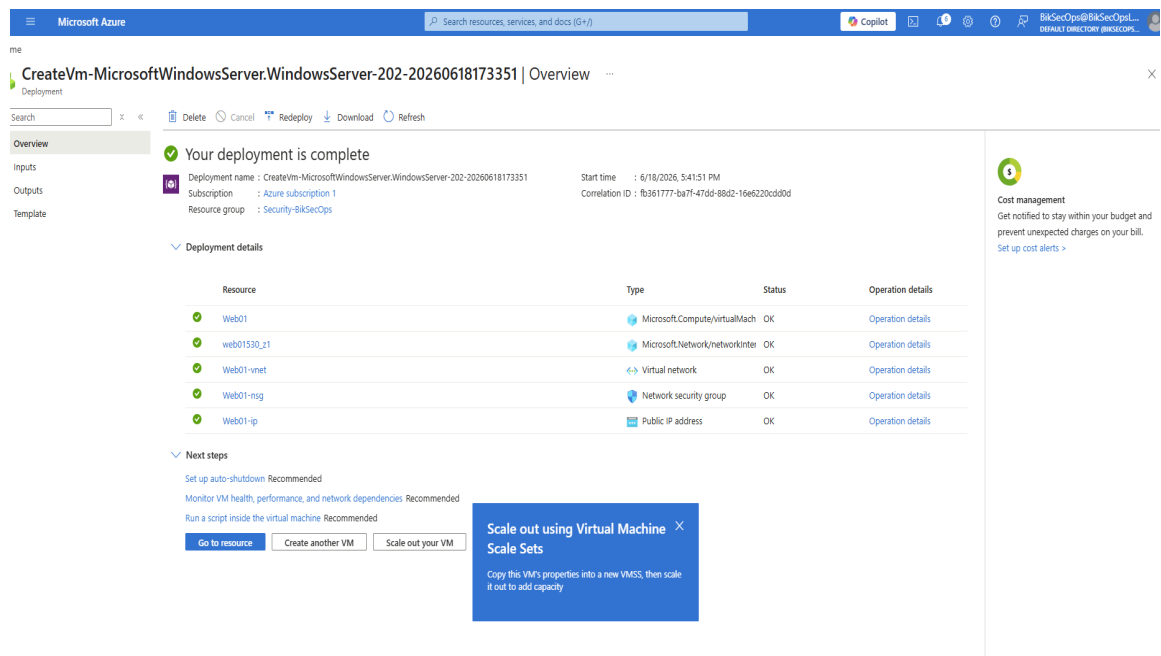


Fig 3 -- Azure: Web01 VM deployment complete -- All resources (VM, NIC, NSG, VNet, Public IP) status OK

4.3 Disable Windows Defender Firewall on Web01

RDP connected to Web01, then navigated to Control Panel > Windows Defender Firewall > Turn Windows Defender Firewall on or off. Firewall was turned OFF for both Private and Public network profiles to ensure Hydra brute force traffic was not blocked.

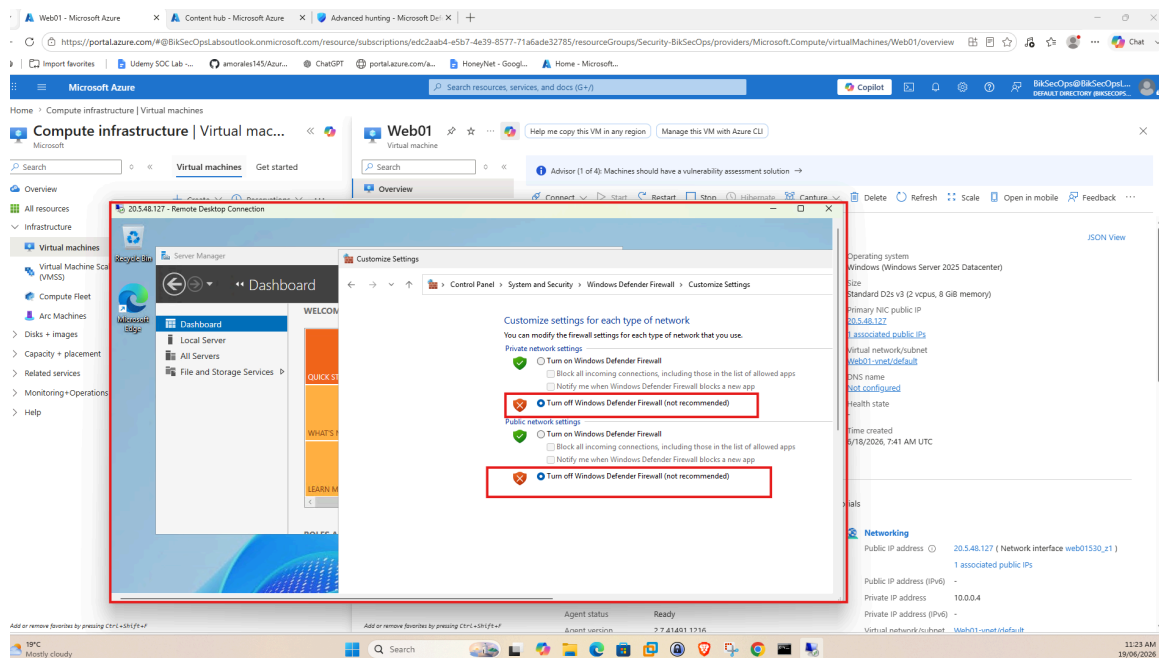


Fig 4 -- Web01 RDP: Windows Defender Firewall disabled for both Private and Public profiles (Turn off selected)

4.4 Create Attacker VM -- AttackerVM (Ubuntu 24.04)

A second VM running Ubuntu 24.04 LTS was deployed in the same resource group to serve as the attack machine.

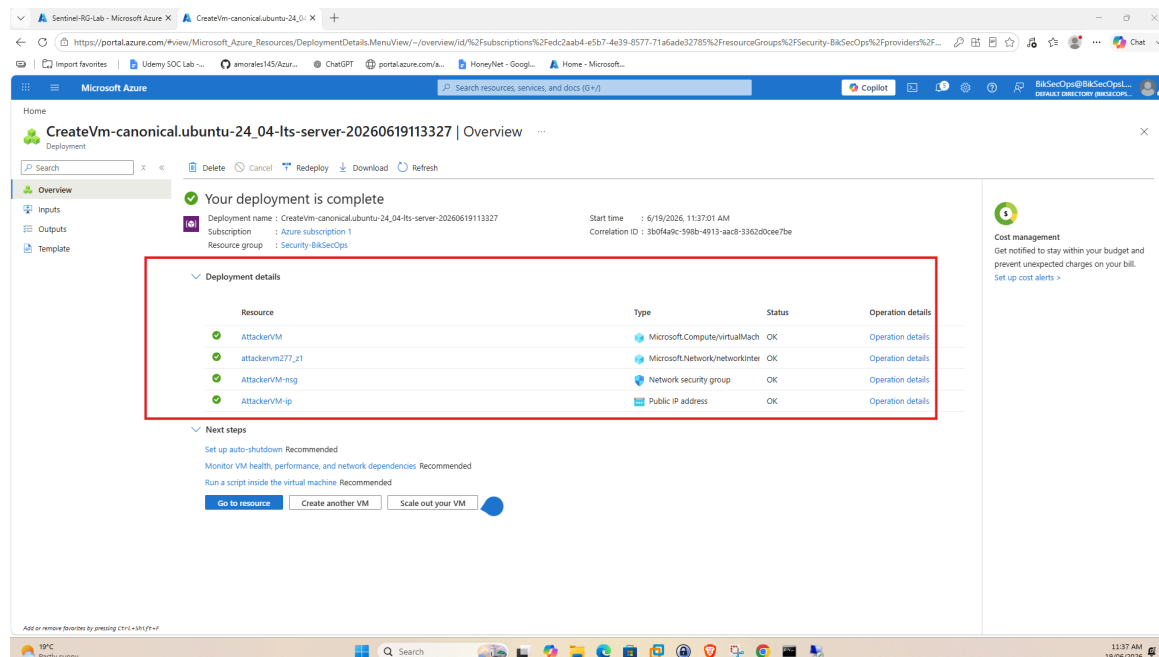


Fig 5 -- Azure: AttackerVM Ubuntu 24.04 deployment complete -- All resources (VM, NIC, NSG, Public IP) status OK

5. Phase 2 -- Log Analytics Workspace & Sentinel Setup

5.1 Create Log Analytics Workspace

Workspace Name	Bik-SOC-Lab
Resource Group	Security-BikSecOps
Region	Australia East

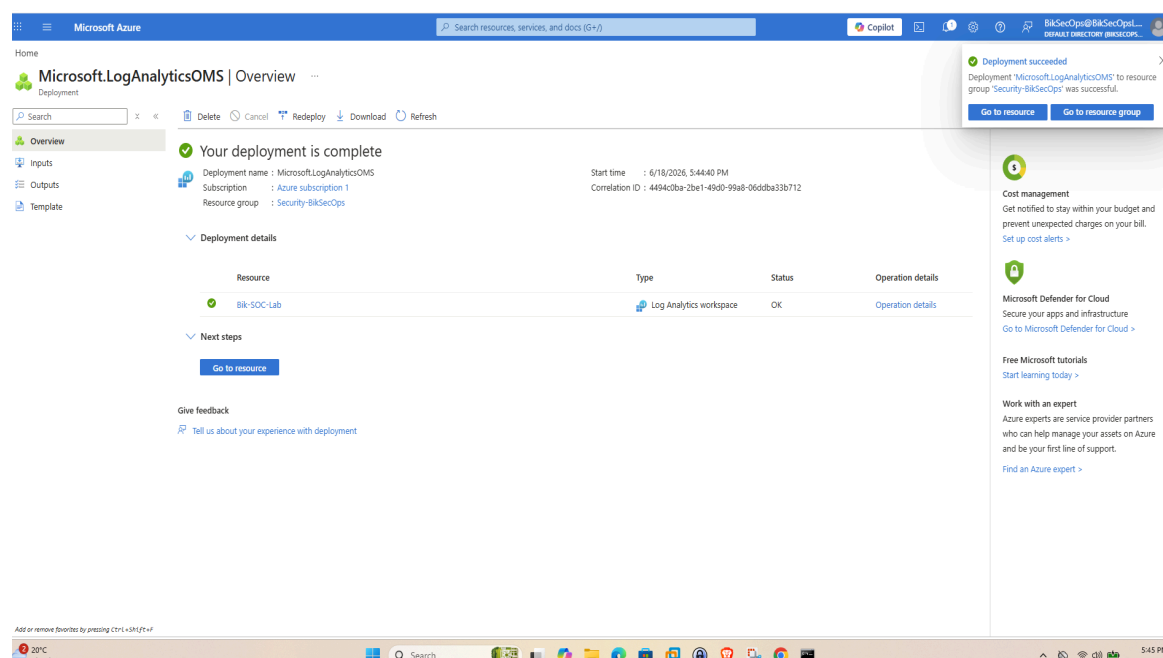


Fig 6 -- Azure: Bik-SOC-Lab Log Analytics Workspace deployment complete

5.2 Add Bik-SOC-Lab to Microsoft Sentinel

Microsoft Sentinel was enabled on the Bik-SOC-Lab workspace via the Add Microsoft Sentinel to a workspace page.

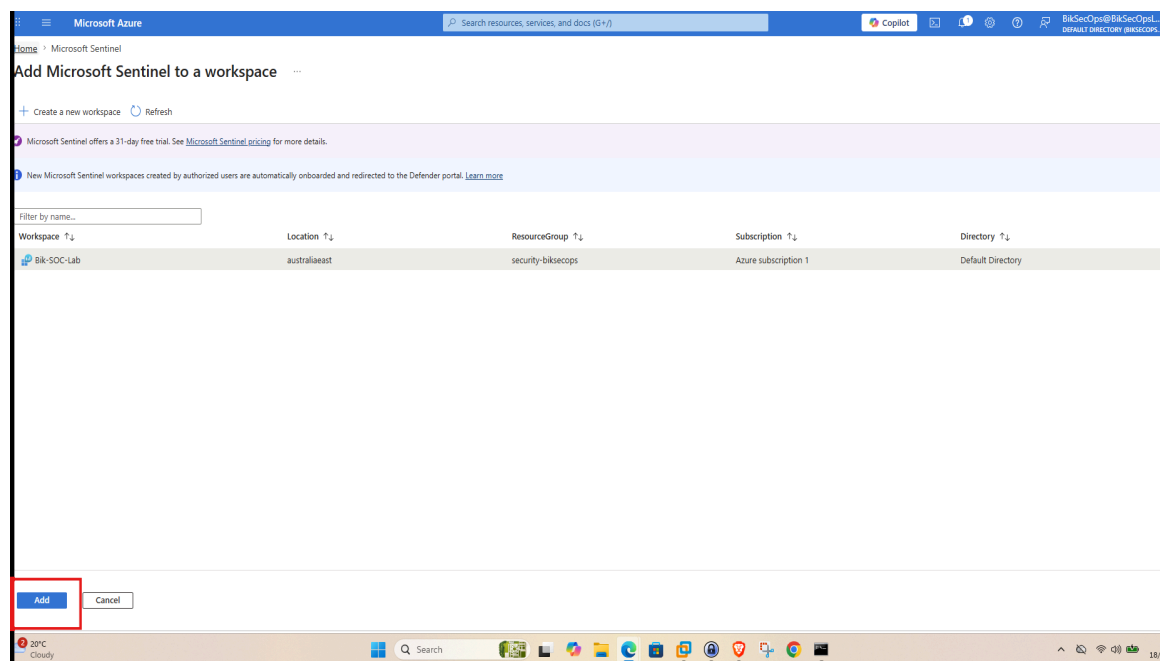


Fig 7 -- Azure: Add Microsoft Sentinel -- Bik-SOC-Lab workspace selected, australiaeast, Security-BikSecOps RG

6. Phase 3 -- Data Connector & Collection Rule

6.1 Install Windows Security Events via AMA Connector

From Sentinel Content Hub, the Windows Security Events solution was installed. This solution provides the Windows Security Events via AMA data connector for ingesting Security Event logs.

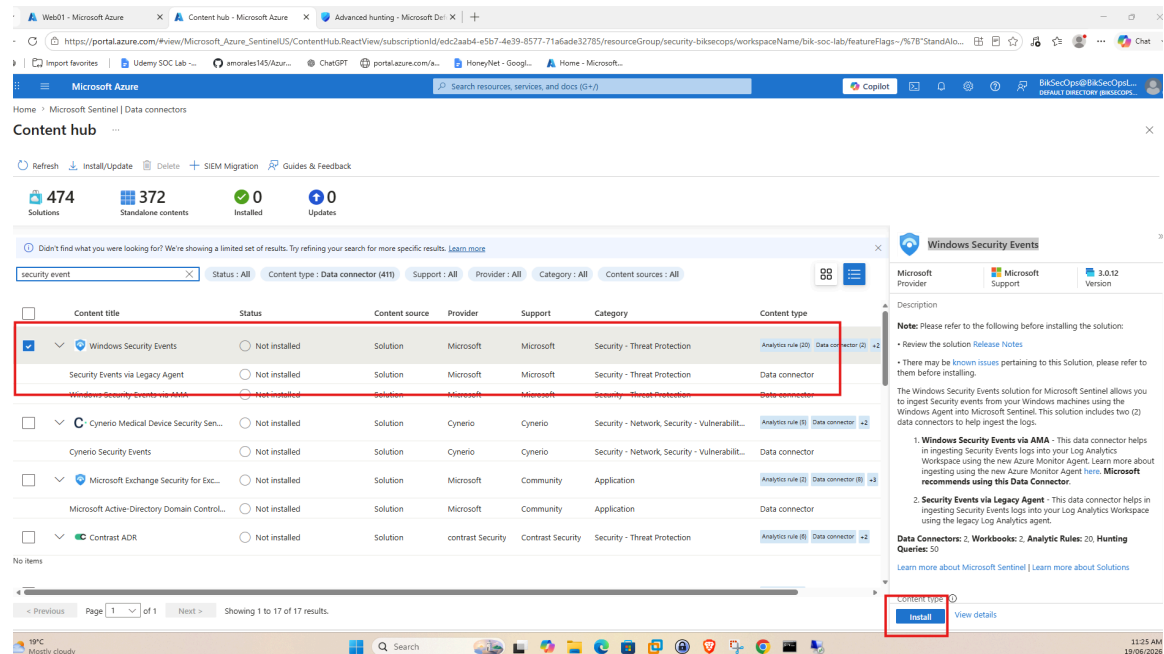


Fig 8 -- Sentinel Content Hub: Windows Security Events selected and installed

6.2 Create Data Collection Rule

A Data Collection Rule was created to collect Windows Security Events from Web01 and send them to the Bik-SOC-Lab workspace.

DCR Name	SecurityEventsforAzureVM
Subscription	Azure Subscription 1
Resource Group	Security-BikSecOps
Selected Resource	Web01 (microsoft.compute/virtualmachines)
Events Collected	All Events

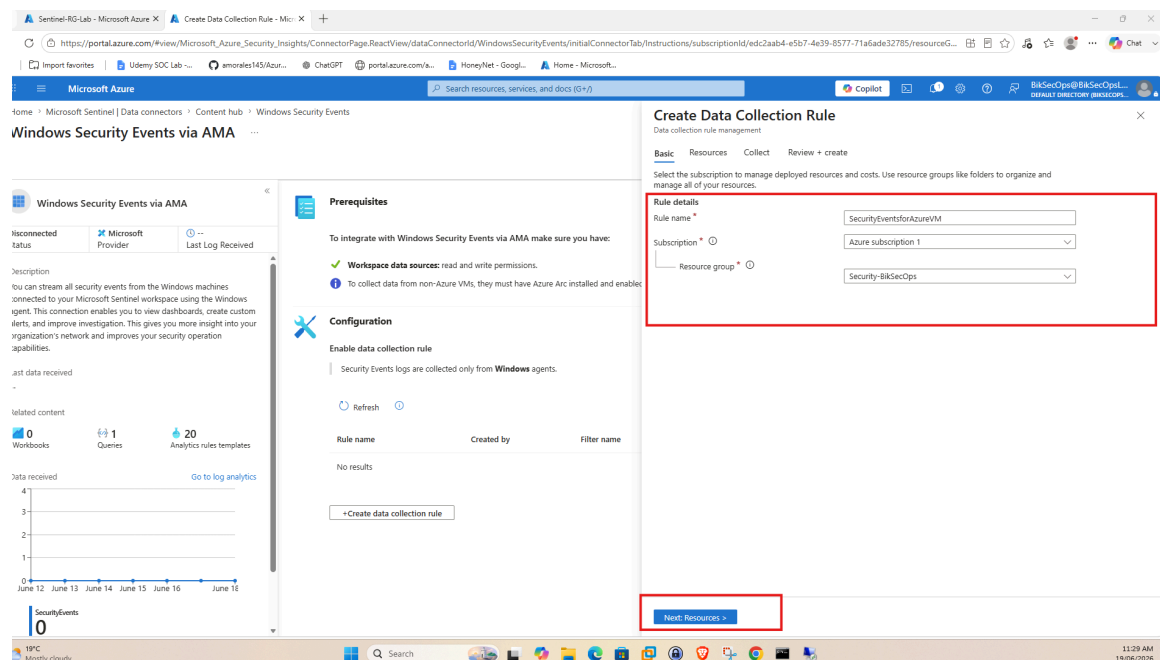


Fig 9 -- Windows Security Events via AMA: Create Data Collection Rule -- SecurityEventsforAzureVM, Web01 resource

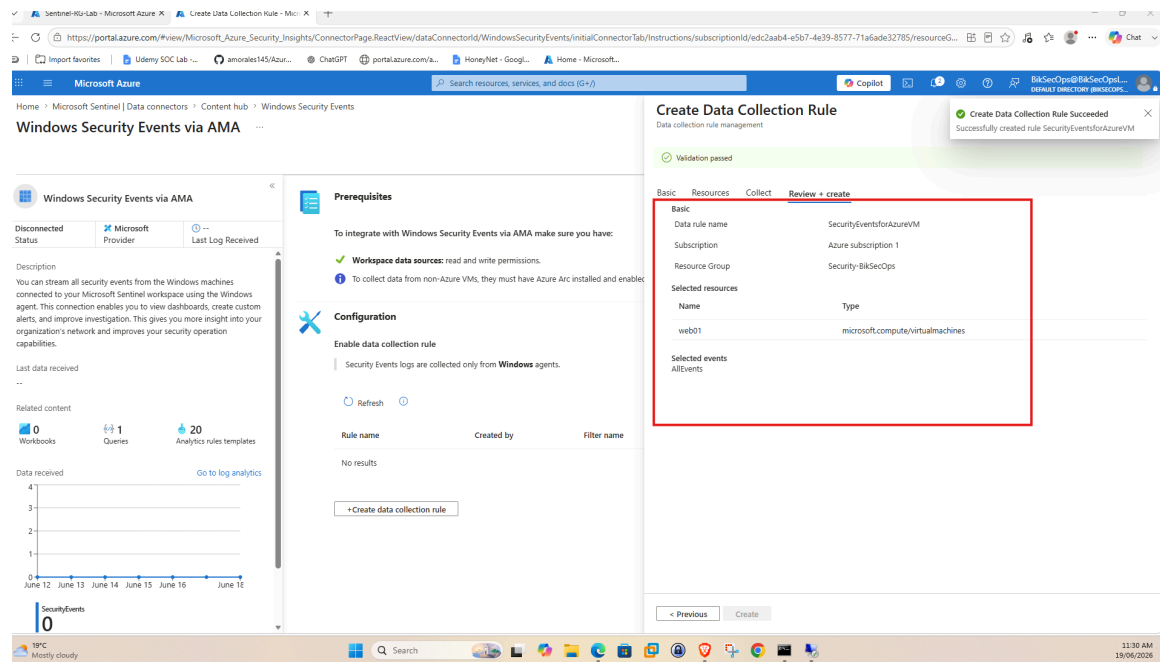


Fig 10 -- DCR Creation Success: SecurityEventsforAzureVM -- Web01 selected, All Events, Validation passed

7. Phase 4 -- Attack Simulation (RDP Brute Force with Hydra)

7.1 Connect to AttackerVM via Azure Bastion

Azure Bastion was used to connect to AttackerVM securely without exposing SSH ports. Authentication type: VM Password, Username: Attackuser.

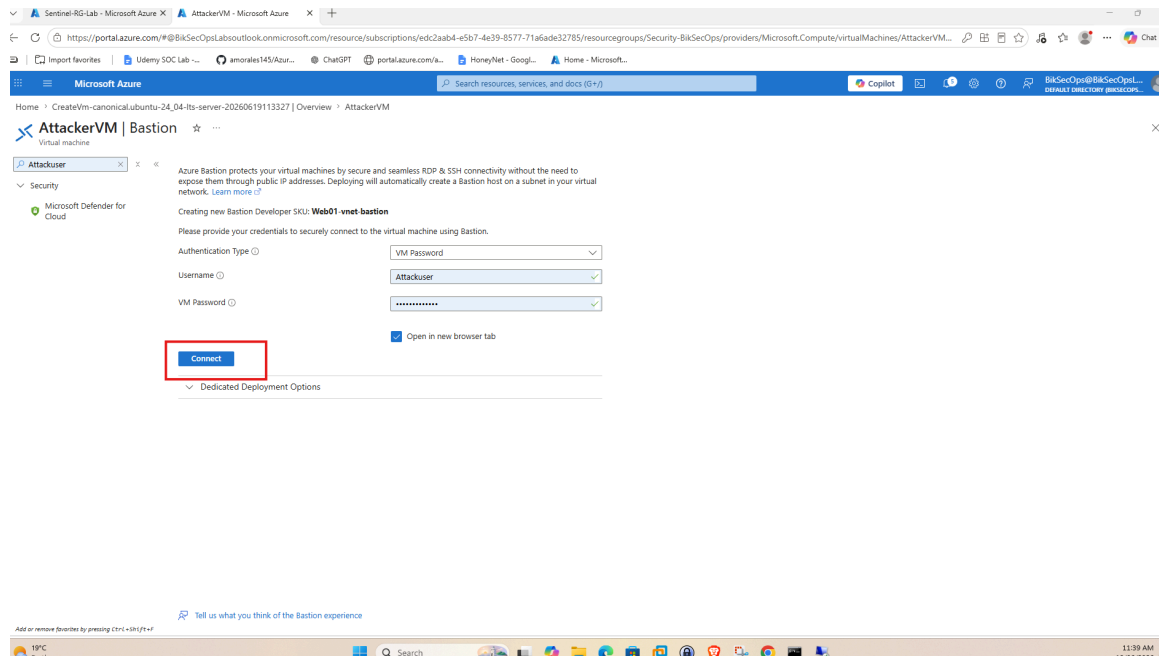


Fig 11 -- Azure Bastion: Connect to AttackerVM -- Authentication type: VM Password, Username: Attackuser

7.2 Update AttackerVM and Install Hydra

```
sudo apt update
sudo apt install hydra
```

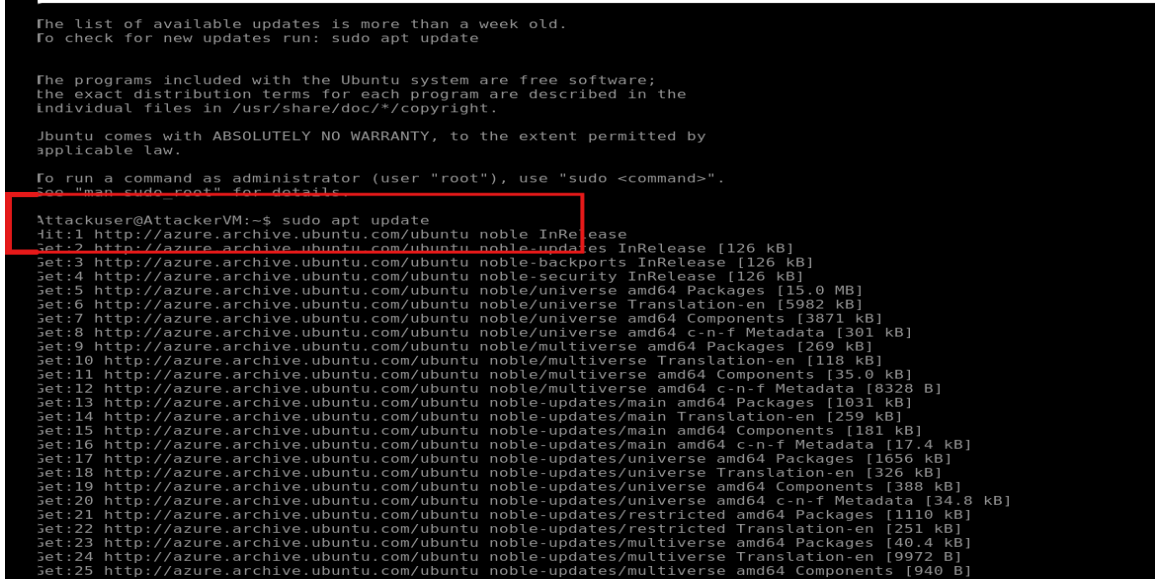


Fig 12 -- AttackerVM Terminal: sudo apt update running -- Ubuntu package lists updating

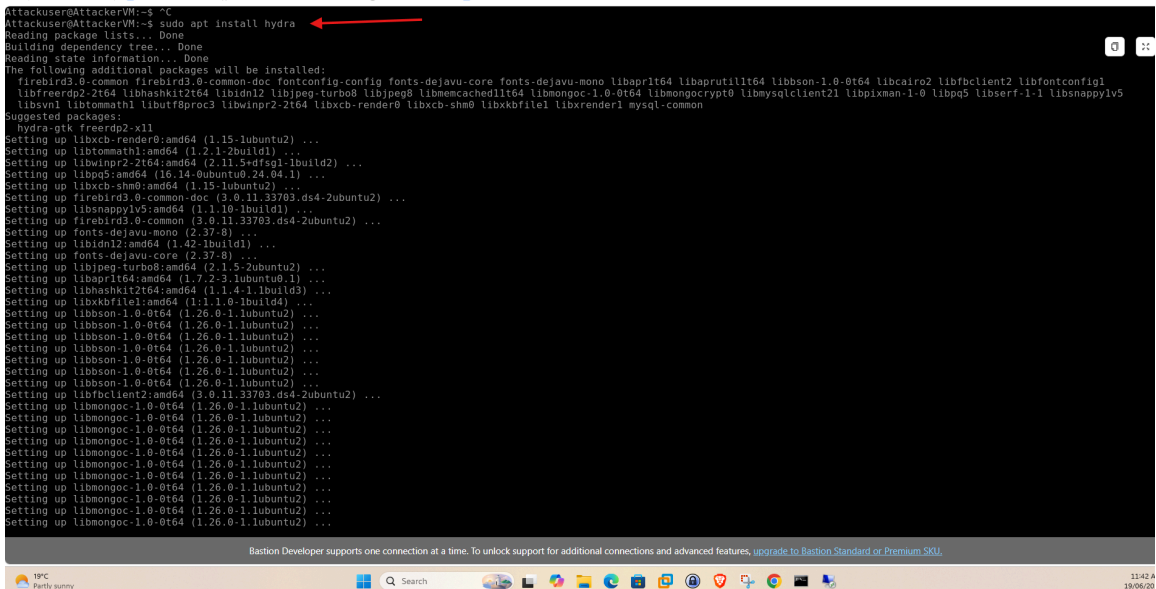


Fig 13 -- AttackerVM Terminal: sudo apt install hydra -- Hydra v9.5 installation complete

7.3 Create Password List and Add Correct Password

A commonly used password list was downloaded from the SecLists GitHub repository, then the actual Web01 password (bikashsoc123) was added to the list using nano to guarantee a successful authentication at the end of the attack.

```
wget
raw.githubusercontent.com/danielmiessler/SecLists/refs/heads/master/Passwords/Common-Credentials/2025-199_most_used_passwords.txt
nano 2025-199_most_used_passwords.txt
```

Added "bikashsoc123" to the password list and saved the file.

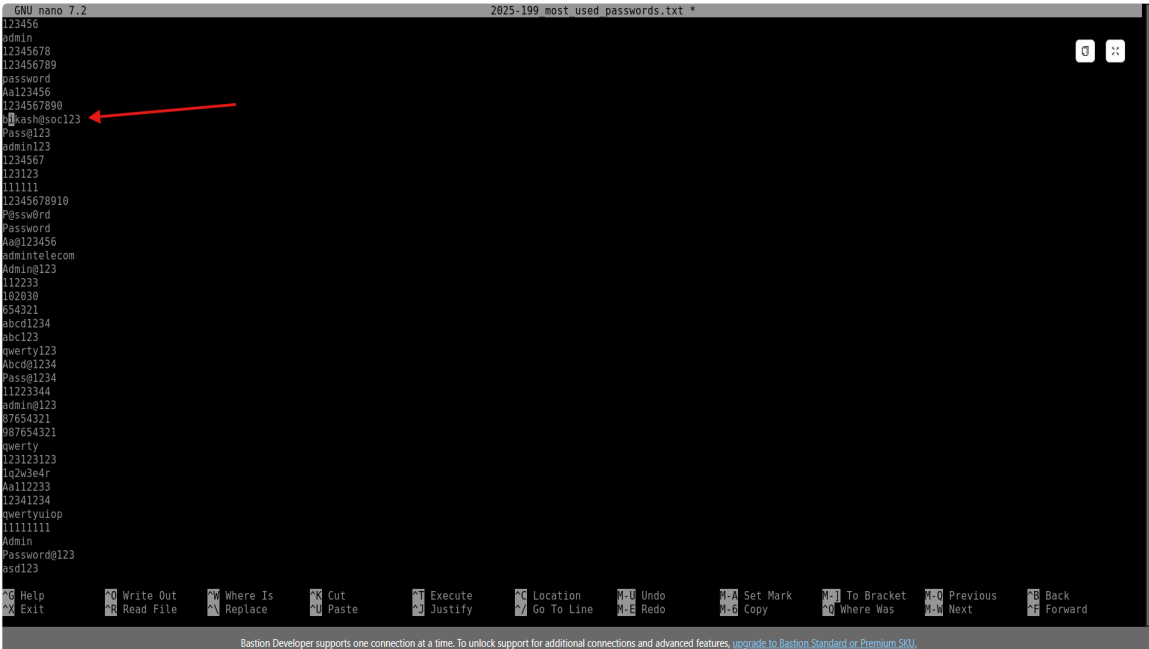


Fig 14 -- AttackerVM nano: Password list open -- bikashsoc123 highlighted (added to list to simulate successful brute force)

7.4 Launch RDP Brute Force Attack

Hydra was used to launch a dictionary-based RDP attack against Web01 public IP on port 3389:

```
hydra -l bikashsoc1 -P 2025-199_most_used_passwords.txt -t 1 -W 5
rdp://20.5.48.127:3389
```

-l bikashsoc1	Target username (Web01 local admin)
-P	Password list file (with correct password added)
-t 1	One task (RDP module is experimental)
-W 5	Wait 5 seconds between attempts
Result	1 of 1 target successfully completed. 1 valid password found: bikash@soc123

```
123123
111111
12345678910
P@ssw0rd
Password
Aa0123456
admin@telecom
Admin@123
112233
182030
054321
abcd1234
abc123
qwerty123
Abcd01234
Pass01234
11223344
admin@123
07654321
007654321
qwerty
123123123
1q2w3e4r
Aa112233
12341234
qwertyuiop
11111111
Admin
Password@123
asd123

Attacker@AttackerVM:~$
Attacker@AttackerVM:~$ hydra -l bikashsoc1 -P 2025-199_most_used_passwords.txt -t 1 -W 5 rdp://20.5.48.127:3389
hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws
and ethics anyway).

hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-22 01:53:23
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.
[DATA] max 1 task per 1 server, overall 1 task, 199 login tries (l:l/p:199), -199 tries per task
[3389][rdp] host: 20.5.48.127 login: bikashsoc1 password: bikash@soc123
1 of 1 target successfully completed, 1 valid password found
hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-06-22 01:54:01
Attacker@AttackerVM:~$
```

Fig 15 -- AttackerVM Hydra: Attack complete -- "1 valid password found" -- bikashsoc1:bikash@soc123

8. Phase 5 -- Log Verification & KQL Threat Hunting

After the Hydra attack completed, Sentinel Logs were queried to confirm that both failed (4625) and successful (4624) logon events were captured for user bikashsoc1.

KQL Query -- Brute Force Followed by Successful Login

```
SecurityEvent
| where TimeGenerated > ago(15m)
| where EventID in (4624, 4625)
| summarize
    FailedAttempts = countif(EventID == 4625),
    SuccessAttempts = countif(EventID == 4624),
    FirstFailure = minif(TimeGenerated, EventID == 4625),
    LastSuccess = maxif(TimeGenerated, EventID == 4624)
by TargetUserName, IPAddress
| where FailedAttempts >= 7
| where SuccessAttempts > 0
| where LastSuccess > FirstFailure
```

The query returned one result row confirming the attack:

TargetUserName	bikashsoc1
IpAddress	20.70.152.193 (AttackerVM public IP)
FailedAttempts	7
SuccessAttempts	1
FirstFailure	6/22/2026, 1:53:24 AM
LastSuccess	6/22/2026, 1:54:00 AM

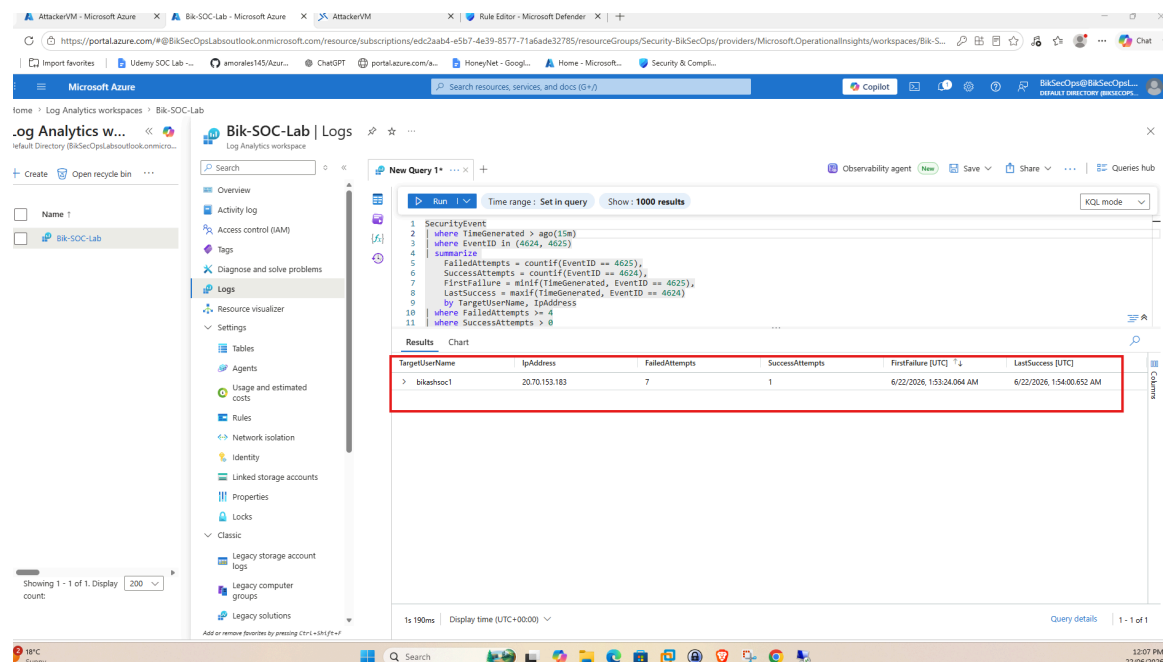


Fig 16 -- Sentinel Logs: KQL query results -- bikashsoc1 from 20.70.152.193: 7 failed + 1 success, LastSuccess > FirstFailure confirmed

9. Phase 6 -- Analytics Detection Rule

A Scheduled Analytics Rule was created in Microsoft Sentinel (via the Defender XDR unified portal) to automatically detect this brute force pattern and generate incidents.

Rule Name	Successful Brute Force Attempt
Description	Detects multiple failed logon attempts followed by a successful logon from the same IP address. This may indicate a brute-force attack, password spraying activity, or successful account compromise.
Severity	High
Status	Enabled
MITRE Tactic	Credential Access Initial Access
MITRE Techniques	T1110 Brute Force T1110.001 Password Guessing T1110.003 Password Spraying T1078 Valid Accounts
Rule Frequency	Every 5 minutes
Lookback Period	Last 5 hours
Trigger Threshold	More than 0 results
Entity Mapping	Account: Name > TargetUserName IP Address: Address > IpAddress

Analytics Rule KQL Query

```
SecurityEvent
| where EventID in (4624, 4625)
| where LogonType in (2, 3, 10)
| summarize
    FailedAttempts = countif(EventID == 4625),
    SuccessAttempts = countif(EventID == 4624),
    LastFailed = maxif(TimeGenerated, EventID == 4625),
    SuccessTime = maxif(TimeGenerated, EventID == 4624)
    by TargetUserName, IpAddress
| where FailedAttempts >= 5
| where SuccessAttempts > 0
| where SuccessTime > LastFailed
| project TargetUserName, IpAddress, FailedAttempts, SuccessAttempts, LastFailed,
SuccessTime
| order by SuccessTime desc
```

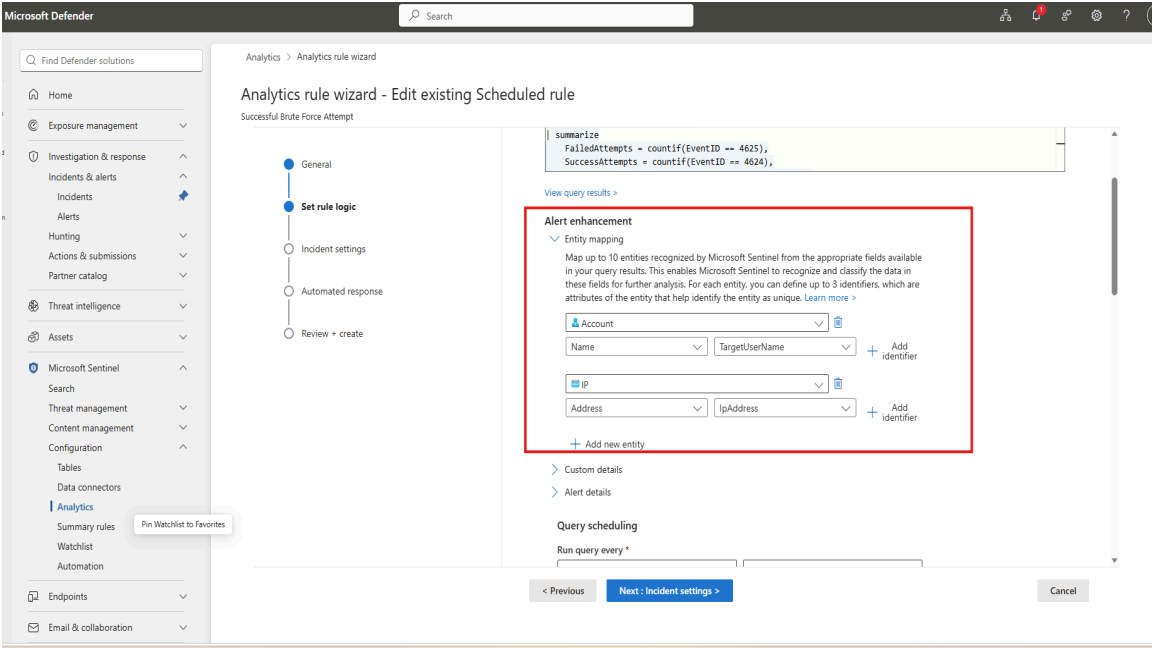


Fig 17 -- Defender XDR Analytics Rule Wizard: Entity mapping -- Account: TargetUserName, IP Address: IpAddress

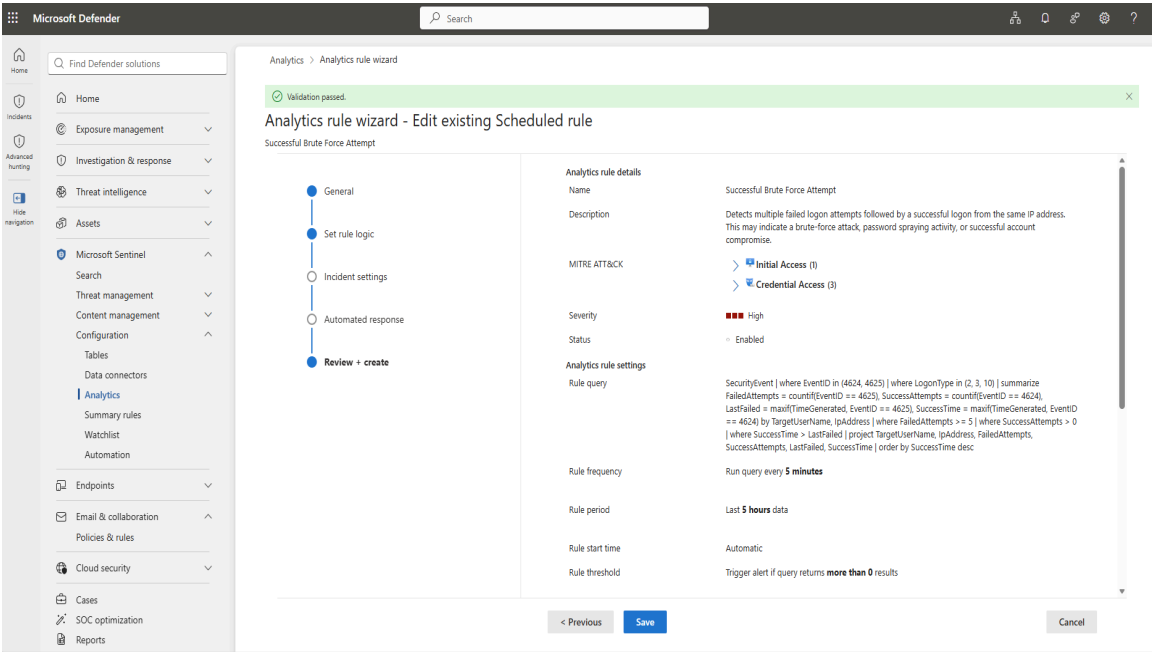


Fig 18 -- Defender XDR: Analytics rule Review+create -- Successful Brute Force Attempt, High severity, Enabled, 5-minute frequency

10. Phase 7 -- Incident Detection & Response

10.1 Incident Triggered in Microsoft Defender XDR

The analytics rule triggered multiple alerts which were grouped into Incident ID 18 in the Microsoft Defender XDR unified portal.

Incident ID	18
Incident Title	Successful Brute Force Attempt
Severity	High
Status	Active
Total Alerts	10
Active Alerts	2/2
Affected Entity	bikashsoc1 (user)
Attack IP	20.70.151.193
First Activity	Jun 22, 2026 7:57 AM
Last Activity	Jun 22, 2026 1:02 PM
Workspace	Bik-SOC-Lab
Classification	Credential access

The screenshot displays the Microsoft Defender XDR console for Incident ID 18, titled "Successful Brute Force Attempt". The interface is divided into several sections:

- Left Sidebar:** Contains navigation links for Home, Incidents, Alerts, Hunting, Advanced hunting, Actions & submissions, Partner catalog, Threat intelligence, Assets, Microsoft Sentinel, Search, Threat management, Content management, Configuration, Tables, Data connectors, Analytics, Summary rules, Watchlist, Automation, and Endpoints.
- Top Bar:** Shows the incident title, severity (High), status (Active), and a search bar.
- Main Content Area:**
 - Attack story:** Displays a list of alerts (2/2) and activities (2). The first activity is "Successful Brute Force Attempt" at Jun 22, 2026 7:57:54 AM, and the last activity is "Successful Brute Force Attempt" at Jun 22, 2026 1:02:54 PM.
 - Incident graph:** A visual representation of the attack path, showing a connection from the IP address 20.70.151.193 to the user entity bikashsoc1.
 - Incident details:** A summary of the incident, including the assigned status (Unassigned), incident ID (18), classification (Credential access), first activity (Jun 22, 2026 7:57:54 AM), last activity (Jun 22, 2026 1:02:54 PM), and workspace (Bik-SOC-Lab).
- Right Sidebar:** Contains a "Priority assessment" section indicating that the incident does not yet have a priority assigned, and a "Description" section detailing the attack path.

Fig 19 -- Microsoft Defender XDR: Incident ID 18 -- Successful Brute Force Attempt -- High severity, Active, bikashsoc1 entity, attack graph showing IP to user

10.2 Create Investigation Task

An investigation task was created on the incident to document the scope of work and guide the response.

Task Name	User / Host Review
Category	Investigate
Status	In Progress
Priority	High
Affected Account	bikashsoc1 (Privileged)
Affected Host	Web01 (Production / External Facing)
Investigation Steps	1. Validate authentication activity 2. Review failed and successful logon events 3. Identify source IP and confirm attack path 4. Determine whether unauthorized access occurred
Containment/Recommended Actions	1. Disable affected account 2. Reset password 3. Revoke active sessions 4. Block source IP 5. Isolate host if compromise confirmed 6. Escalate to DFIR if malicious

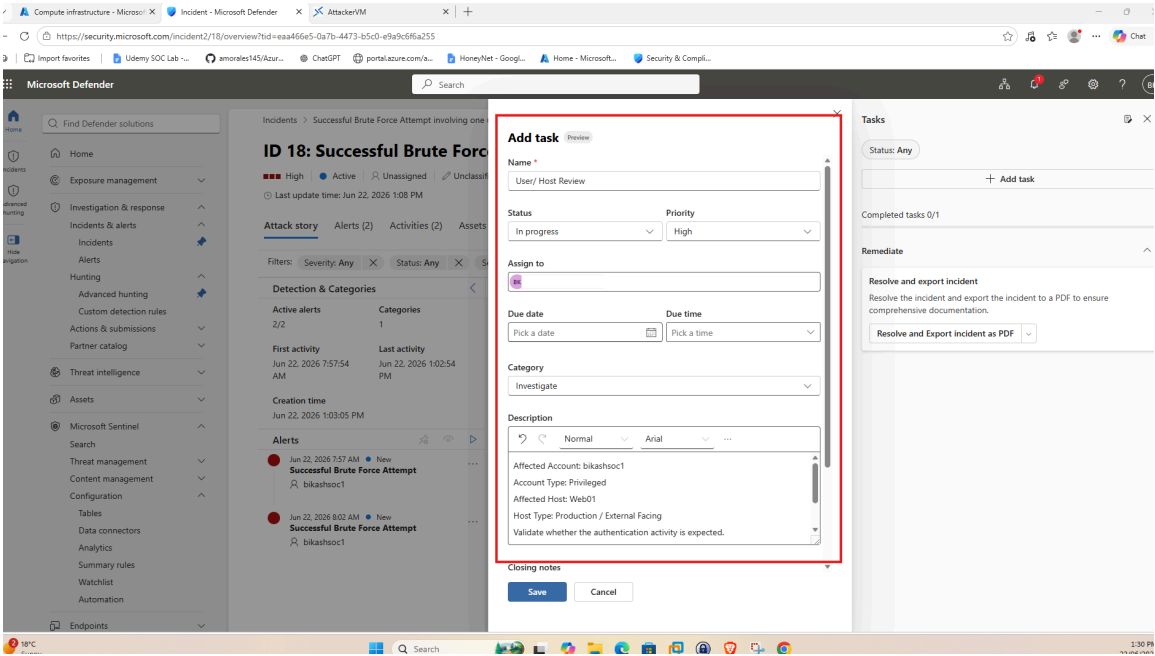


Fig 20 -- User/Host Review, Category: Investigate, bikashsoc1, Affected Host: Web01, containment actions documented

10.3 Incident Resolution & Closing Notes

After investigation confirmed this was authorized security testing activity, the incident was resolved and closed with a detailed closing note.

Final Status	Resolved
Assigned To	BikSecOps@BiksecOpsLabsoutlook.onmicrosoft.com
Classification	Informational: expected activity -- Confirmed activity
Closing Note	As this activity was intentionally generated for detection validation purposes, no malicious activity was identified. Incident closed as Expected Activity / Security Testing.
Tasks Completed	2/2

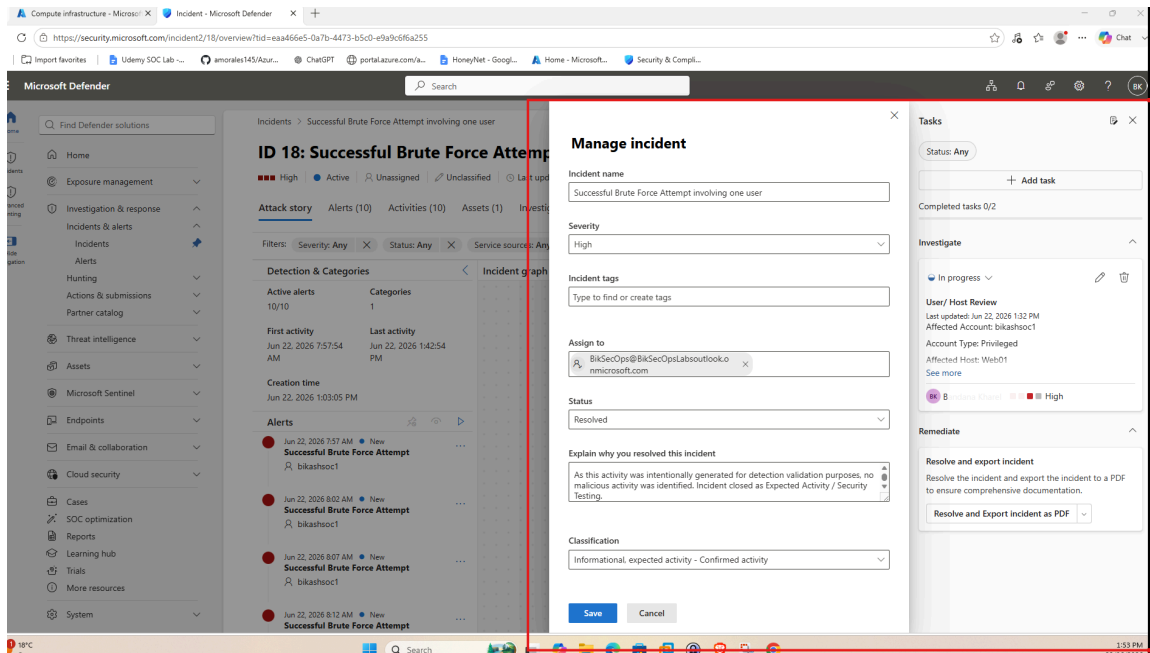


Fig 21 -- Defender XDR: Manage incident -- Status: Resolved, Classification: Informational expected activity, Closing notes added, Tasks 2/2 completed

10.4 Simulated Production Response (If Activity Was Malicious)

Had this been confirmed as a genuine attack, the following actions would have been taken:

Immediate Containment:

- Disable bikashsoc1 account immediately to prevent further unauthorized access
- Block attacker IP 20.70.152.193 at NSG level with explicit Deny rule
- Revoke all active RDP sessions on Web01
- Isolate Web01 from the network to prevent lateral movement

Eradication:

- Reset compromised credentials for bikashsoc1 with strong password and enforce MFA
- Re-enable Windows Defender Firewall on Web01 for both Public and Private profiles
- Replace permissive NSG rule (Any/Any/Allow) with IP-restricted RDP access to known management IPs only

- Scan for persistence mechanisms including scheduled tasks, new local accounts, and registry run keys

Recovery:

- Restore Web01 from pre-attack snapshot if system integrity was compromised
- Monitor Web01 authentication logs for 7 days post-recovery for any recurrence
- Re-enable bikashsoc1 account only after full verification and password reset confirmed

Escalation:

- Escalate to DFIR team if lateral movement or data exfiltration was suspected
- Notify security leadership and document full incident timeline for post-incident review

11. Skills Demonstrated

Microsoft Sentinel Deployment	Azure NSG & Firewall Configuration	Log Analytics Workspace Setup
Windows Security Events via AMA	Data Collection Rule (DCR) Creation	KQL Threat Hunting & Detection
Custom Analytics Rule Creation	MITRE ATT&CK Framework Mapping	Incident Detection & Triage
Microsoft Defender XDR Portal	Incident Response Procedures	RDP Brute Force Simulation (Hydra)
Linux CLI (Ubuntu / Bastion)	SOC Workflow End-to-End	Security Event Log Analysis

12. Conclusion

This lab successfully demonstrated a complete end-to-end SOC workflow using Microsoft Sentinel and the Microsoft Defender XDR unified portal. Starting from raw Azure infrastructure, the lab covered vulnerable VM configuration, log ingestion pipeline setup via AMA connector and DCR, real attack simulation using Hydra RDP brute force, KQL-based threat detection confirming 7 failed logins followed by a successful authentication, and a custom high-severity analytics rule creation mapped to MITRE ATT&CK T1110.

The analytics rule successfully triggered Incident ID 18 in the Defender XDR portal. The incident was triaged, an investigation task was created documenting the affected account (bikashsoc1), affected host (Web01), source IP (20.70.152.193), and recommended containment steps. The incident was closed as Expected Activity / Security Testing after confirming the activity was generated for detection validation.

References

Microsoft Sentinel Documentation -- <https://learn.microsoft.com/azure/sentinel/>
Microsoft Defender XDR -- <https://security.microsoft.com>
MITRE ATT&CK T1110 Brute Force -- <https://attack.mitre.org/techniques/T1110/>
THC-Hydra (Brute Force Tool) -- <https://github.com/vanhauser-thc/thc-hydra>
SecLists Common Passwords -- <https://github.com/danielmiessler/SecLists>
KQL Reference -- <https://learn.microsoft.com/azure/data-explorer/kql-quick-reference>